Assignment- 3

2303A51672

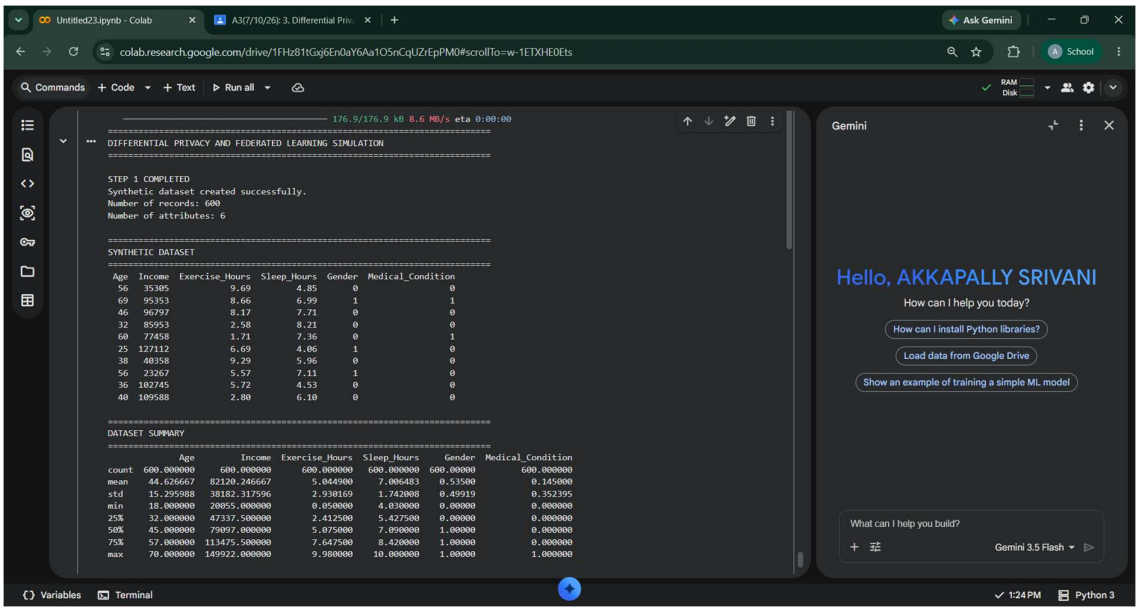
Batch- 23

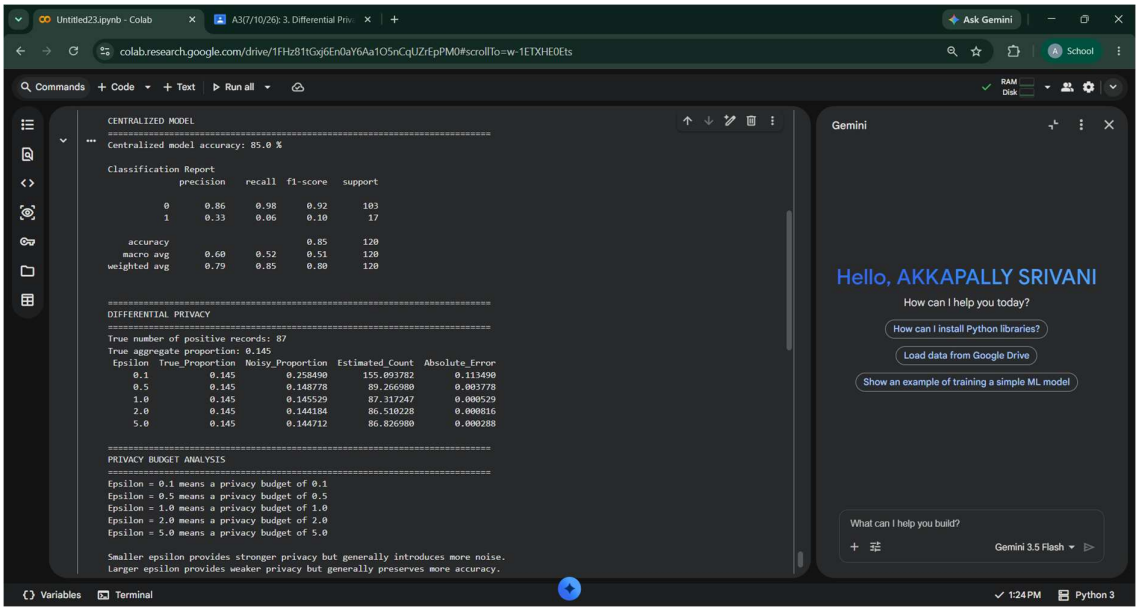
Differential Privacy and Federated Learning – Conceptual/ Simulation
Scenario:
Federated Learning – Conceptual/ Simulation

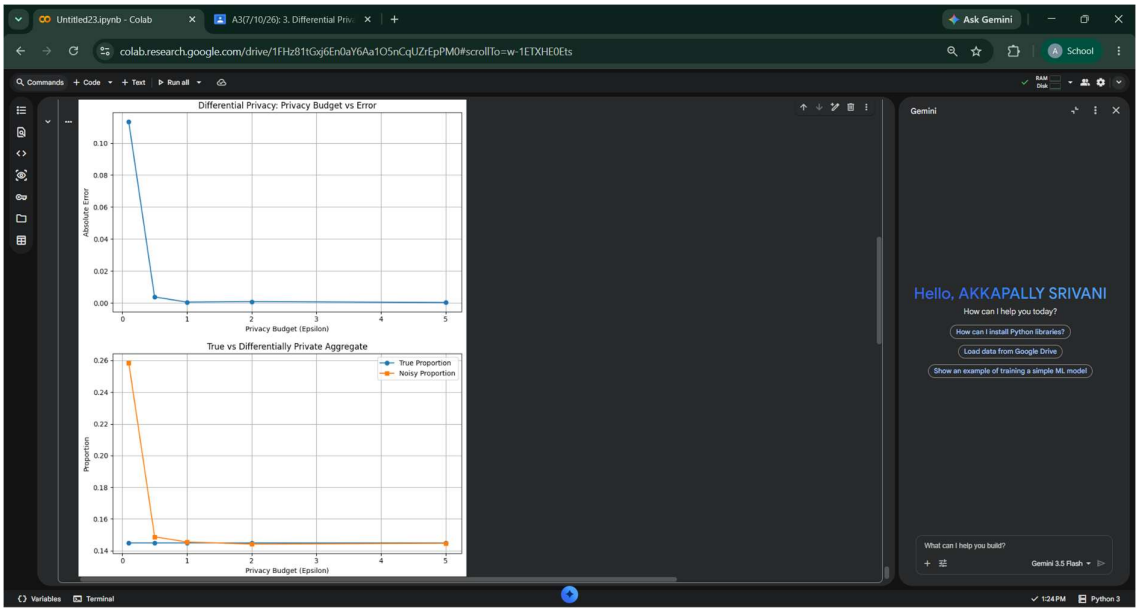
Task:
• Implement or simulate a simple differential-privacy mechanism using an approved library/framework.
• Observe how controlled noise affects aggregate statistics or model utility.
• Explain the role of privacy budget conceptually.
• Study a simple federated-learning workflow using local client data and aggregated updates, or simulate the process.
• Compare centralized data sharing with distributed training.
• Document benefits, limitations and assumptions.

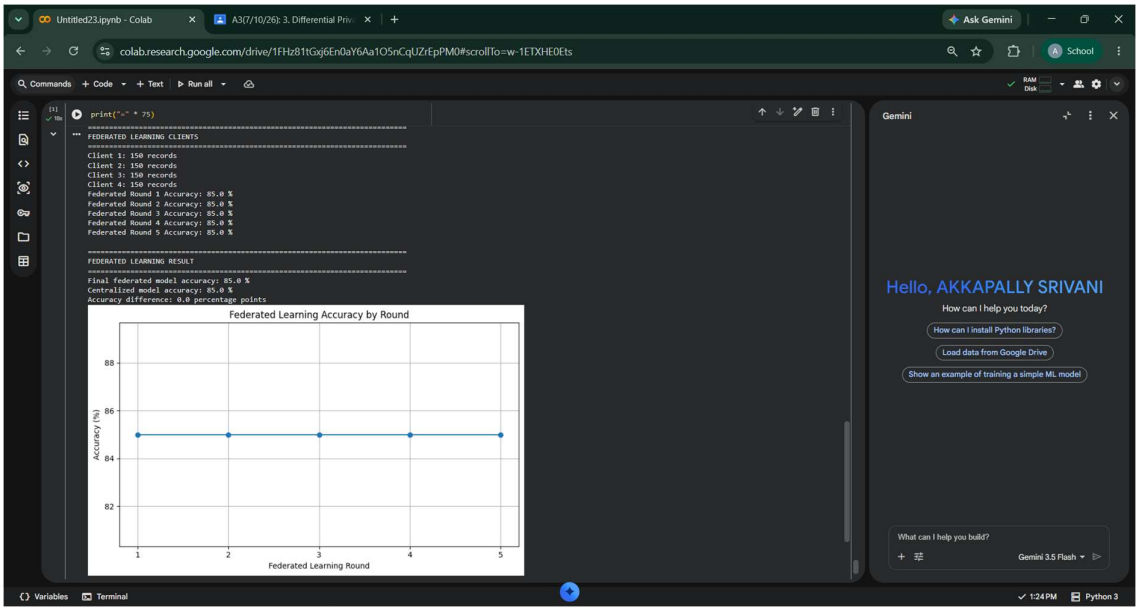
Focus:
Comparing privacy-preserving learning concepts and their trade-offs.

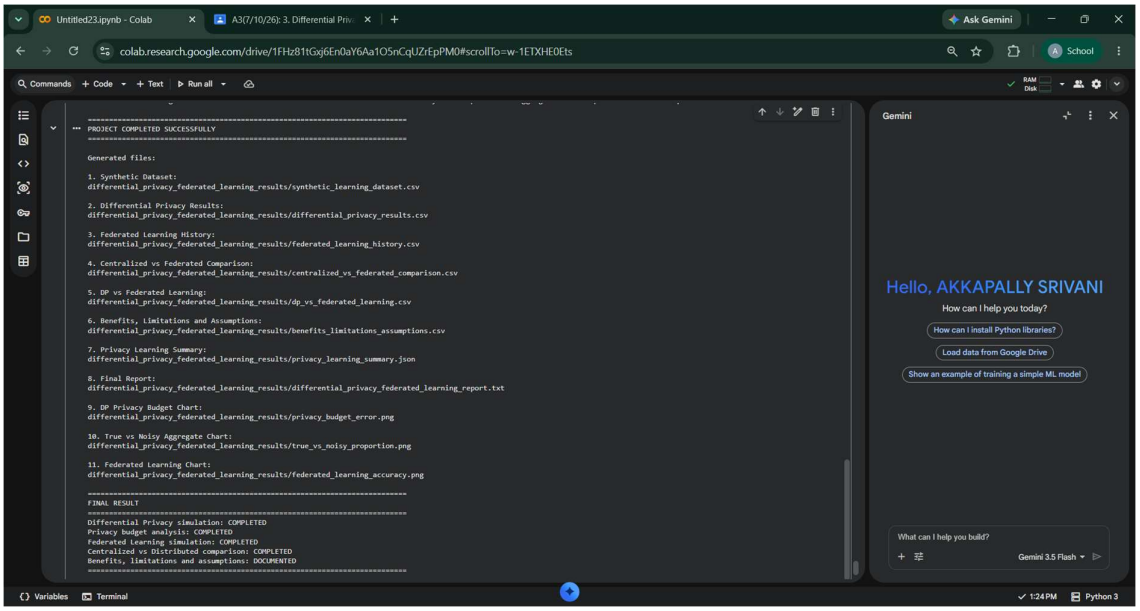
Output:




Untitled23.ipynb - Colab
A3(7/10/26): 3. Differential Priv
Ask Gemini
colab.research.google.com/drive/1FHz81tGxj6En0aY6Aa1O5nCqUZrEpPM0#scrollTo=w-1ETXHE0Ets
School
Commands
+ Code
+ Text
Run all
RAM
Disk
CENTRALIZED MODEL
==========
Centralized model accuracy: 85.0 %
Classification Report
precision recall f1-score support
0 0.86 0.98 0.92 103
1 0.33 0.06 0.10 17
accuracy 0.85 120
macro avg 0.60 0.52 0.51 120
weighted avg 0.79 0.85 0.80 120
==========
DIFFERENTIAL PRIVACY
==========
True number of positive records: 87
True aggregate proportion: 0.145
Epsilon True_Proportion Noisy_Proportion Estimated_Count Absolute_Error
0.1 0.145 0.258490 155.093782 0.113490
0.5 0.145 0.148778 89.266980 0.003778
1.0 0.145 0.145529 87.317247 0.000529
2.0 0.145 0.144184 86.510228 0.000816
5.0 0.145 0.144712 86.826980 0.000288
==========
PRIVACY BUDGET ANALYSIS
==========
Epsilon = 0.1 means a privacy budget of 0.1
Epsilon = 0.5 means a privacy budget of 0.5
Epsilon = 1.0 means a privacy budget of 1.0
Epsilon = 2.0 means a privacy budget of 2.0
Epsilon = 5.0 means a privacy budget of 5.0
Smaller epsilon provides stronger privacy but generally introduces more noise.
Larger epsilon provides weaker privacy but generally preserves more accuracy.
Gemini
Hello, AKKAPALLY SRIVANI
How can I help you today?
How can I install Python libraries?
Load data from Google Drive
Show an example of training a simple ML model
What can I help you build?
Gemini 3.5 Flash
Variables
Terminal
1:24 PM
Python 3


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Ask Gemini
colab.research.google.com/drive/1FHz81tGxj6En0aY6Aa1O5nCqUZrEpPM0#scrollTo=w-1ETXHE0Ets
School
Commands
+ Code
+ Text
Run all
Differential Privacy: Privacy Budget vs Error
Absolute Error
Privacy Budget (Epsilon)
True vs Differentially Private Aggregate
True Proportion
Noisy Proportion
Proportion
Privacy Budget (Epsilon)
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School
Commands
Code
Text
Run all
RAM
Disk
print("=" * 75)
FEDERATED LEARNING CLIENTS
Client 1: 150 records
Client 2: 150 records
Client 3: 150 records
Client 4: 150 records
Federated Round 1 Accuracy: 85.0 %
Federated Round 2 Accuracy: 85.0 %
Federated Round 3 Accuracy: 85.0 %
Federated Round 4 Accuracy: 85.0 %
Federated Round 5 Accuracy: 85.0 %
FEDERATED LEARNING RESULT
Final federated model accuracy: 85.0 %
Centralized model accuracy: 85.0 %
Accuracy difference: 0.0 percentage points
Federated Learning Accuracy by Round
Accuracy (%)
88
86
84
82
1
2
3
4
5
Federated Learning Round
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Commands
Code
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Run all
RAM
Disk
PROJECT COMPLETED SUCCESSFULLY
Generated files:
1. Synthetic Dataset:
differential_privacy_federated_learning_results/synthetic_learning_dataset.csv
2. Differential Privacy Results:
differential_privacy_federated_learning_results/differential_privacy_results.csv
3. Federated Learning History:
differential_privacy_federated_learning_results/federated_learning_history.csv
4. Centralized vs Federated Comparison:
differential_privacy_federated_learning_results/centralized_vs_federated_comparison.csv
5. DP vs Federated Learning:
differential_privacy_federated_learning_results/dp_vs_federated_learning.csv
6. Benefits, Limitations and Assumptions:
differential_privacy_federated_learning_results/benefits_limitations_assumptions.csv
7. Privacy Learning Summary:
differential_privacy_federated_learning_results/privacy_learning_summary.json
8. Final Report:
differential_privacy_federated_learning_results/differential_privacy_federated_learning_report.txt
9. DP Privacy Budget Chart:
differential_privacy_federated_learning_results/privacy_budget_error.png
10. True vs Noisy Aggregate Chart:
differential_privacy_federated_learning_results/true_vs_noisy_proportion.png
11. Federated Learning Chart:
differential_privacy_federated_learning_results/federated_learning_accuracy.png
FINAL RESULT
Differential Privacy simulation: COMPLETED
Privacy budget analysis: COMPLETED
Federated Learning simulation: COMPLETED
Centralized vs Distributed comparison: COMPLETED
Benefits, limitations and assumptions: DOCUMENTED
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